

Exercise I: Constituency Parsing

Given Sentence “**She ate the cake on the table.**” and the following Context-Free Grammar (CFG):

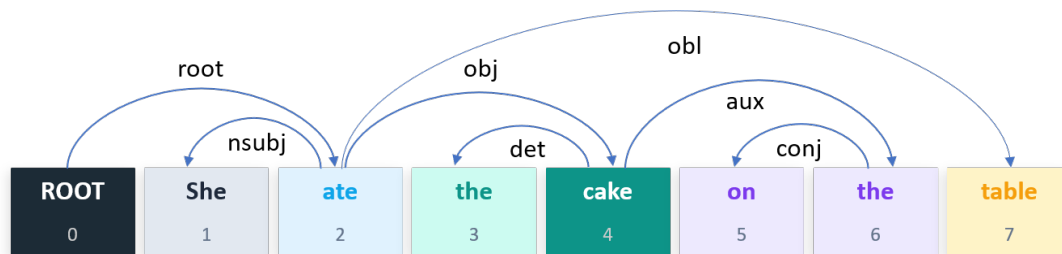
$S \rightarrow NP VP$ $NP \rightarrow Det N \mid NP PP \mid N$ $VP \rightarrow V NP \mid VP PP$ $PP \rightarrow P NP$	$Det \rightarrow the$ $N \rightarrow She \mid cake \mid table$ $V \rightarrow ate$ $P \rightarrow on$
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Identify the terminals in CFG.

- Starting from the start symbol S, derive the sentence step by step. Provide two different derivations corresponding to the two possible interpretations of the sentence.
- Draw two constituency trees corresponding to your derivations.
- Explain the two different meanings of the sentence based on the two parse trees.

Exercise II: Dependency Parsing

Given Sentence “**She ate the cake on the table**” and the following Dependency Tree that contains errors:



- Apply a transition-based parser and fill in the steps (Stack, Buffer, Action, Dependency).
- Compute UAS and LAS using the incorrect tree as Prediction and the following correct tree as Gold.

